08/01/2025

Code-A_Phase-1

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : Aakash Tower, 8, Pusa Road, New Delhi-110005, *Ph.*011-47623456

MM : 720

NCERT Booster Test Series for NEET (RM)-2024-25_T03A

Time : 200 Min.

Topics Covered:
Physics: Electrostatic Potential and Capacitance, Current Electricity, Moving Charges and Magnetism, Magnetism and Matter, Electromagnetic Induction, Alternating Current, Electromagnetic Waves, Ray Optics & Optical Instruments
Chemistry: Chemical Kinetics, Organic Chemistry: Basic Principles & Techniques, Hydrocarbons, Haloalkanes and Haloarenes , Alcohols, Phenols and Ethers, Aldehydes, Ketones and Carboxylic Acids, Amines (Organic Compound containing Nitrogen)
Botany: Photosynthesis in Higher Plants, Respiration in Plants, Plant Growth and Development, Sexual Reproduction in Flowering Plants, Principles of Inheritance & Variation, Molecular Basis of Inheritance
Zoology: Animal Kingdom-II: Non-chordates (Annelida to Hemichordata), Chordates, Structural Organisation in Animals– Animal Morphology (Cockroach, Frog, Earthworm), Human Reproduction, Reproductive Health

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks.** For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

SECTION-A

1. Ohm's law is valid if

   (1) V is directly proportional to I^3

   (2) The relation V and I is non-linear

   (3) V is directly proportional to I^2

   (4) I depends on V linearly

2. A dielectric slab of dielectric constant $K = 2$ is completely inserted in the space between a parallel plate capacitor of capacitance 10 μF. The capacitance with dielectric is

   (1) 5 μF

   (2) 10 μF

   (3) 15 μF

   (4) 20 μF

3. For a point source of electromagnetic wave, intensity (I) at a distance r from it, vary as

   (1) $I \propto r$

   (2) $I \propto r^0$

   (3) $I \propto \frac{1}{r}$

   (4) $I \propto \frac{1}{r^2}$

4. A small coil of of N turn has area A and a current 'i' flows through it. The Magnetic dipole moment of the coil

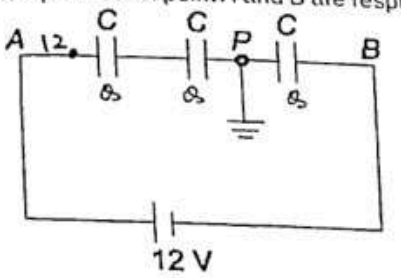
(1) iNA

(2) i^2NA

(3) iN^2A

(4) $\frac{iN}{A}$

5. Three identical capacitors are connected in series as shown in figure, across a 12 V battery. Point P is earthed. The potential of point A and B are respectively



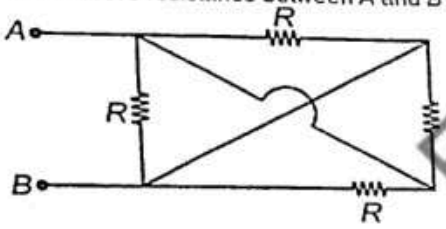
(1) 12 V, 0 V

(2) −6 V, 6 V

(3) 8 V, −4 V

(4) −4 V, 8 V

6. The effective resistance between A and B will be



(1) $\frac{R}{3}$

(2) $\frac{R}{4}$

(3) 2R

(4) 3R

7. Electric charge Q is distributed non-uniformly over a thin ring of radius r, then potential V at a point on the axis of the ring at a distance x from the centre of the ring is

(1) $V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{(r^2+x^2)}$

(2) $V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{\sqrt{r^2+x^2}}$

(3) $V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{(r^2+x^2)^{\frac{1}{2}}}$

(4) Information is not sufficient

8. Energy of 2 eV in joule is equal to

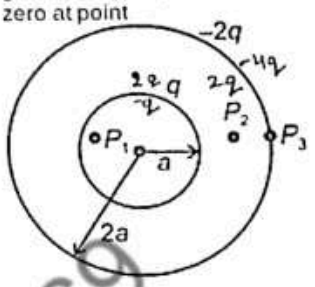
(1) 8.2×10^{-19} J

(2) 3.2×10^{-19} J

(3) 1.6×10^{-19} J

(4) 4.8×10^{19} J

9. Two concentric conducting shells of radius a and 2a are given charges q and −2q respectively. The potential will be zero at point



(1) P_1

(2) P_2

(3) P_3

(4) Both P_1 and P_3

10. A charged particle moving in a magnetic field $\vec{B} = (\hat{i} - \hat{j})T$, has an acceleration of $(2\hat{i} + \alpha\hat{j})$ m/s^2 at some instant. Find α.

(1) 1

(2) 2

(3) 3

(4) 4

11. Which one of the following statement is incorrect?

(1) A magnet does not exert a force on a stationary electrically charged body.

(2) Magnet field lines are closed continuous curves and extend from south pole to north pole within the magnet.

(3) The tangent to the magnetic field line at any point gives the direction of the magnetic field at that point.

(4) Magnetic field lines are widely separated where the magnetic field is strong.

12. The current passing through a choke coil of 5 Henry is decreasing at the rate of 2 A s^{-1}. The emf developing across the coil is

(1) 10 V

(2) 5 V

(3) 2.5 V

(4) 20 V

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13. Two inductor coils L_1 = 18 mH and L_2 = 8 mH are perfectly coupled. If a current of 2 A is reversed in first coil within 2 millisecond, then the EMF induced in the second coil will be

(1) 12 volt

(2) 24 volt

(3) 12 mV

(4) 24 mV

14. Choose the wrong statement for electromagnetic wave. They

(1) Are transverse in nature

(2) Travel in vacuum with the speed of light c

(3) Are produced by accelerated charges

(4) Travel with same speed in all mediums

15. The Maxwell's equations $\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i + \varepsilon_0 \frac{d\phi_E}{dt} \right)$ is a statement of

(1) Faraday's law of EMI

(2) Modified Ampere's law

(3) Gauss' law of electrostatic

(4) Gauss' law of magnetism

16. Electrostatic potential in a region varies with coordinates x, y, z as $V = (3x - 2y + 4z)$ volt. The electric field in this region is

(1) Uniform

(2) Non-uniform varying with x

(3) Non-uniform varying with x and y both

(4) Non-uniform varying with x, y and z

17. If the value of permeability of a nickel-iron alloy is 0.126 T m A^{-1}, then the susceptibility is approximately

(1) 0.5×10^5

(2) 10^5

(3) 1.5×10^5

(4) 2×10^5

18. Curie's temperature is the temperature

(1) Above which ferromagnetic substance becomes a paramagnetic substance

(2) Below which paramagnetic substance becomes diamagnetic substance

(3) Below which diamagnetic substance becomes superconductor

(4) Above which a superconductor becomes semiconductor

19. The energy stored in an inductor is 2 mJ when a current of 0.2 A is established in it. The self-inductance of the coil is

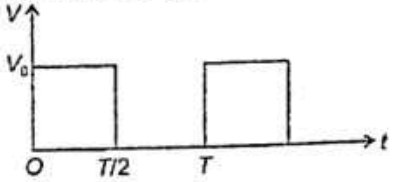
(1) 10 H

(2) 2 H

(3) 0.1 H

(4) 20 H

20. The r.m.s. value of potential difference V shown in the figure between 0 and $\frac{T}{2}$ is



(1) V_0

(2) $\frac{V_0}{2}$

(3) $\frac{V_0}{\sqrt{2}}$

(4) $\frac{V_0}{3}$

21. **Assertion (A):** A pure inductor acts as perfect conductor in steady state DC circuit.
Reason (R): For steady state in DC circuit, inductor behaves as open circuit.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) Both (A) and (R) are incorrect

22. A convex lens of focal length 50 cm forms the image of a distant object which subtends an angle of 2 milliradian at the lens. The height of image is

(1) 50 mm

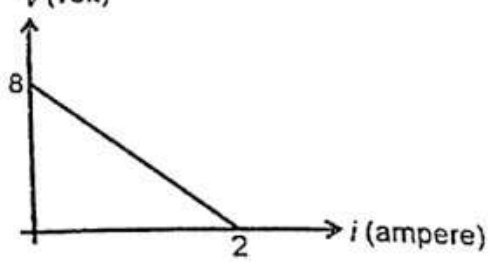
(2) 1 mm

(3) 2 mm

(4) 0.1 mm

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23. A battery of emf E and internal resistance r is connected across a variable load resistance R. A graph is plotted between the current in the circuit (i) and terminal potential difference of the battery.

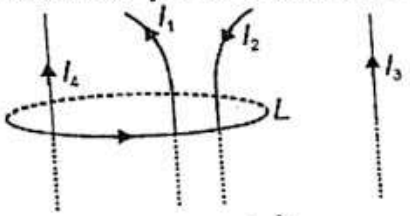


Based on the above information, match the entries in column I and column II. All the parameters in column II are in SI units.

Column I		Column II
A.	Emf of the battery	(P) Zero
B.	Internal resistance of the battery	(Q) 2
C.	Maximum current derived through the battery	(R) 8
D.	Value of load resistance when maximum current is derived	(S) 4

(1) A(R), B(S), C(Q), D(P)
(2) A(R), B(S), C(P), D(Q)
(3) A(P), B(Q), C(R), D(S)
(4) A(Q), B(R), C(S), D(P)

24. The value of $\oint \vec{B} \cdot \vec{dl}$ for the loop L shown in figure is



(1) $\mu_0 (I_1 + I_2 - I_4)$
(2) $\mu_0 (I_1 + I_2 - I_3 + I_4)$
(3) $\mu_0 (I_1 - I_2 + I_4)$
(4) $\mu_0 (I_1 + I_4 - I_3)$

25. A plane electromagnetic wave propagating in vacuum along y-direction has a wavelength of 6 mm. The electric field in x-direction has amplitude of 30 V m^{-1}. The amplitude of magnetic field is given by

(1) 10^{-4} T
(2) 10^{-3} T
(3) 10^{-7} T
(4) 10^{-10} T

26. In a simple microscope, if the final image is located at infinity then its magnifying power is (Where symbols have their usual meaning)

(1) $1 + \frac{D}{f}$
(2) $1 + \frac{f}{D}$
(3) $\frac{D}{f}$
(4) $\frac{f}{D}$

27. An ideal transformer with 20 secondary turns, step down the input voltage from 220 V AC to 22 V AC. The number of turns in the primary coil is

(1) 20
(2) 2
(3) 2000
(4) 200

28. An L-R circuit has cell of e.m.f E, which is switched ON at time $t = 0$, the current in the coil after long time will be

(1) Zero
(2) $\frac{E}{\sqrt{L^2+R^2}}$
(3) $\frac{E}{L}$
(4) $\frac{E}{R}$

29. Magnetic flux due to a magnetic field $2\hat{T}j$, through an area $4\ m^2$ $(\hat{i}+\hat{j})$ is

(1) 16 Wb
(2) 8 Wb
(3) 4 Wb
(4) 2 Wb

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30. If the charge flowing through the cross-section of a wire is given as :
$Q(t) = (2t^2 - 4t)$ C, then current in wire at $t = 2$ s is

(1) 4 A
(2) Zero
(3) 2 A
(4) 8 A

31. A short bar magnet is placed with its axis 30° with an external magnetic field of 0.8 T, experiences a torque of 0.016 N m. How much work is required to turn it to 90° with magnetic field from this position?

(1) 0.028 J
(2) 0.073 J
(3) 0.06 J
(4) 0.35 J

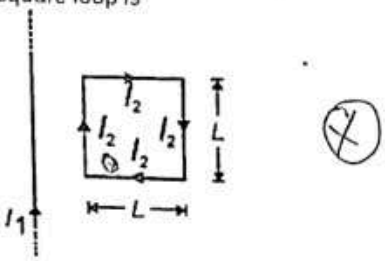
32. The magnification of a compound microscope is 42. If the focal length of eye-piece is 5 cm and the image is formed at least distance of distinct vision, then the magnification of objective lens is

(1) 6
(2) 9
(3) 7
(4) 3

33. A proton and a deuteron are accelerated from rest upto same kinetic energy. If a uniform magnetic field is set up perpendicular to their velocities, the ratio of the radii of the circular paths will be

(1) 1 : 2
(2) 1 : 4
(3) 1 : $\sqrt{2}$
(4) 1 : 2 $\sqrt{2}$

34. In the situation shown in figure, the magnetic force on the square loop is



(1) Towards left
(2) Towards right
(3) Downwards
(4) Upwards

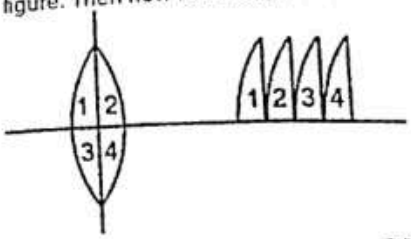
35. For small angled prisms, (whose refracting angle (A) is less than 5°), the correct relation between refractive index (μ) of material of prism and angle of deviation is

(1) $\delta = A(\mu - 1)$
(2) $\delta = \mu A$
(3) $\delta = A(\mu + 1)$
(4) $\delta = A(\mu^2 - 1)$

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SECTION-B

36. A lens is cut from the optical axis into four equal parts. All parts of the lens are placed sideways as shown in the figure. Then new focal length of the combination



(1) Remains the same

(2) Becomes half

(3) Becomes $\frac{1}{4}$th times

(4) Becomes $\frac{1}{8}$th times

37. A charged particle moves in a magnetic field $\vec{B} = 10\hat{i}$ with initial velocity $\vec{u} = 5\hat{i} + 4\hat{j}$. The path of the particle will be

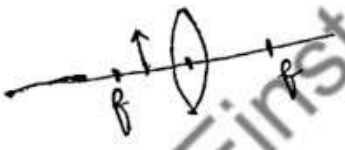
(1) Straight line

(2) Circle

(3) Helical

(4) None

38. An object is placed at a distance $\left(\frac{f}{2}\right)$ from a convex lens. The image will be



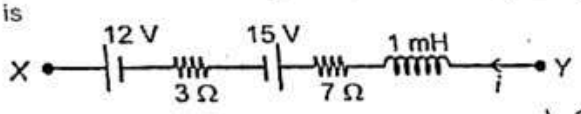
(1) At $\frac{3f}{2}$, real

(2) At f, virtual

(3) At 2f, virtual

(4) At 4f, real

39. The figure shows a given circuit. At the instant, the current i = 2 A and it is decreasing at a rate 2×10^3 A/s, then $V_X - V_Y$ is



(1) 11 V

(2) 21 V

(3) −11 V

(4) −21 V

40. An alternating voltage is connected across a series combination a resistance R and an inductance L If the potential drop across the resistance is 150 V and across the inductance is 150 V, then the applied voltage is approximately

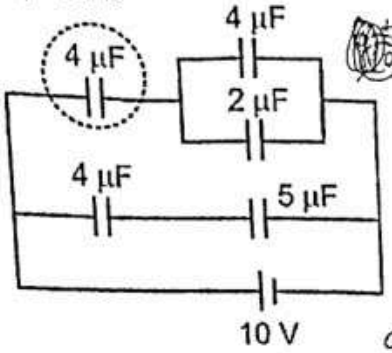
(1) 350 V

(2) 212 V

(3) 500 V

(4) 300 V

41. In the given circuit, the charge on 4 μF connected in figure capacitor is



(1) 36 μC

(2) 24 μC

(3) 16 μC

(4) 90 μC

42. Two equal resistance when connected in series to a battery, consume electric power of 100 W. If these resistance are now connected in parallel combination to the same battery, then electric power dissipated will be

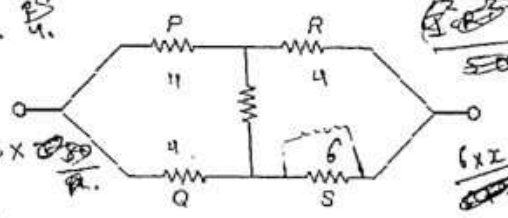
(1) 800 W

(2) 25 W

(3) 200 W

(4) 400 W

43. In Wheatstone's network, $P = 4\ \Omega$, $Q = 4\ \Omega$, $R = 4\ \Omega$ and $S = 6\ \Omega$. The resistance with which S should be shunted in order that the bridge may be balanced is



(1) 4 Ω

(2) 6 Ω

(3) 8 Ω

(4) 12 Ω

44. The refracting angle of prism is 60° and the refractive index of material of prism is $\sqrt{3}$. The angle of minimum deviation is

(1) 180°

(2) 45°

(3) 30°

(4) 60°

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45. For an electric field 2 N/C, a current density 0.5 A/m^2 is produced in a conducting material. The conductivity of the material is

(1) 4 Ω^{-1} m^{-1}

(2) 8 Ω^{-1} m^{-1}

(3) 0.25 Ω^{-1} m^{-1}

(4) 0.5 Ω^{-1} m^{-1}

46. A uniform magnetic field exists in the region given by $\vec{B} = \left(3\hat{i} + 4\hat{j} + 5\hat{k}\right)$ T. A rod of length 5 m placed along y-axis is moved along x-axis with constant speed 2 m s^{-1}. Then induced e.m.f. across the ends of the rod is

(1) Zero

(2) 50 V

(3) 5 V

(4) 10 V

47. Relative permittivity and permeability of material are ε_r and μ_r respectively. Which of the following values of these quantities are allowed for a dimagnetic material?

(1) $\varepsilon_r = 0.5, \mu_r = 1.5$

(2) $\varepsilon_r = 1.5, \mu_r = 0.99$

(3) $\varepsilon_r = 0.5, \mu_r = 0.99$

(4) $\varepsilon_r = 1.5, \mu_r = 1.5$

48. In an ac circuit $I = 100 \sin 200 \pi t$. The time required for the current to achieve its peak value will be

(1) $\frac{1}{100}$ sec

(2) $\frac{1}{200}$ sec

(3) $\frac{1}{300}$ sec

(4) $\frac{1}{400}$ sec

49. A charge of 2 C is moving in a magnetic field of 1 T with velocity 5 m/s. Force experienced by it may be

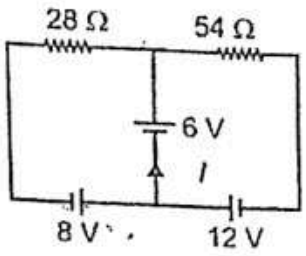
(1) 10 N

(2) 5 N

(3) Zero

(4) All of these

50. Consider the circuit shown in the figure. The current I is equal to



(1) $\frac{6}{5}$ A

(2) 2 A

(3) 3 A

(4) $\frac{5}{6}$ A

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CHEMISTRY

SECTION-A

51. The rate of a reaction get doubled with every 10° rise in temperature. If the temperature of the reaction mixture is raised by 50°C, the new rate of reaction will be

(1) 16 times

(2) 32 times

(3) 64 times

(4) 128 times

52. For an elementary reaction, A + 2B → 2C + D the molecularity is

(1) Zero

(2) One

(3) Two

(4) Three

53. If for a first order reaction $t_{90\%}$ is 50 s then the value of $t_{99.9\%}$ will be

(1) 150 s

(2) 100 s

(3) 10 s

(4) 50 s

54. Time required for 100% completion of a zero order reaction is ($[A]_0$ is the initial concentration of reactant)

(1) $\frac{[A]_0}{k}$

(2) $k[A]_0$

(3) $\frac{k}{[A]_0}$

(4) $2k[A]_0$

55. All natural radioactive decay of unstable nuclei take place follows

(1) Zero order kinetics

(2) First order kinetics

(3) Second order kinetics

(4) Third order kinetics

56. The most stable carbocation, among the following, is

(1) $CH_3-\overset{+}{C}H_2-CH_3$

(2) $(CH_3)_3C-\overset{+}{C}H-CH_3$

(3) $CH_3-CH_2-\overset{+}{C}H-CH_2-CH_3$

(4) $CH_3-\overset{+}{C}H-CH_2-CH_2-CH_3$

57. In Dumas method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is

(1) 15.76

(2) 17.36

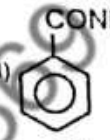
(3) 18.20

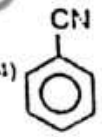
(4) 16.76

58. Kjeldahl method is not applicable for

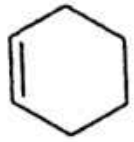
(1) 

(2) 

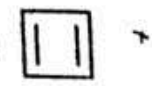
(3) 

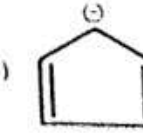
(4) 

59. Aromatic species among the following is

(1) 

(2) 

(3) 

(4) 

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60. Which among the following is not a nucleophile?

(1) H_2O

(2) RNH_2

(3) $AlCl_3$

(4) CH_3O^-

61. Which of the following alkyne can react with Tollen's reagent?

(1) But-2-yne

(2) Pent 2-yne

(3) Hex-3-yne

(4) But-1-yne

62. The decreasing order of boiling points of the following alkanes is

(a) heptane
(b) butane
(c) 2-methylbutane
(d) 2-methylpropane
(e) hexane

Choose the correct answer from the options given below :

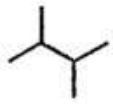
(1) (a) > (e) > (c) > (b) > (d)

(2) (a) > (c) > (e) > (d) > (b)

(3) (c) > (d) > (a) > (e) > (b)

(4) (a) > (e) > (b) > (c) > (d)

63. The compound which is not prepared in good yield by Wurtz reaction is

(1) 

(2) 

(3) 

(4) 

64. When ethyne is heated with mercuric sulphate and dilute H_2SO_4 at 333 K, the product obtained is

(1) Acetone

(2) Acetic anhydride

(3) Acetaldehyde

(4) Formic acid

65. Number of mono chloro derivatives obtained on chlorination of methyl cyclohexane (excluding stereoisomers) is

(1) 3

(2) 5

(3) 6

(4) 8

66. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : $(CH_3)_3C - Cl$ undergoes hydrolysis reaction by S_N1 mechanism.

Reason (R) : $(CH_3)_3\overset{\oplus}{C}$ is very stable carbonium ion.

In the light of above statements, select the **correct** option.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

67. In general the ease of dehydrohalogenation of alkyl halide with alcoholic KOH is

(1) 3° < 2° < 1°

(2) 3° > 2° > 1°

(3) 3° < 2° > 1°

(4) 3° > 2° < 1°

68. Swart's reaction is used to prepare

(1) Alkyl fluorides

(2) Alkyl chlorides

(3) Alkyl bromides

(4) Alkyl iodides

69. Consider the following reaction

$A + B \xrightarrow{H_2SO_4} DDT$

A and B respectively are

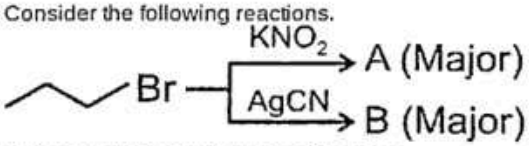
(1) Chlorobenzene and Acetaldehyde

(2) Chlorobenzene and Chloral

(3) Chlorobenzene and Chloroform

(4) Chlorobenzene and Benzaldehyde

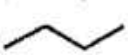
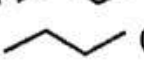
70. Consider the following reactions.



Major products A and B respectively are

(1)  NO_2 and  NC

(2)  ONO and  CN

(3)  NO_2 and  CN

(4) ONO and NC

EinsteinX369

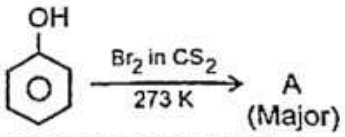
71. Given below are two statements one is labelled as assertion (A) other is labelled as reason (R)
Assertion (A) : Phenol is more acidic than ethanol.
Reason (R) : Phenoxide ion is more stable than ethoxide ion.
In the light of above statements choose the correct answer.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

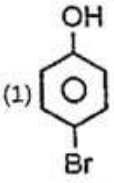
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

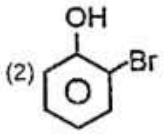
(3) Assertion is true statement but Reason is false

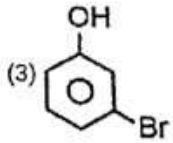
(4) Both Assertion and Reason are false statements

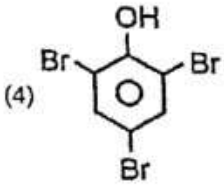
72. 

A is major product. Identify A.

(1) 

(2) 

(3) 

(4) 

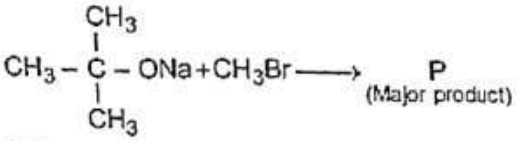
73. The reaction of Lucas reagent is fastest with

(1) CH_3—OH

(2) CH_3CH_2OH

(3) $(CH_3)_2$ CHOH

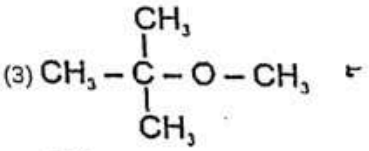
(4) $(CH_3)_3COH$

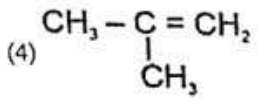
74. 

Major product (P) in the above reaction is

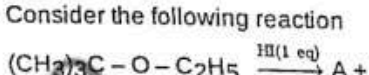
(1) CH_3OH

(2) CH_3OCH_3

(3) 

(4) 

75. Consider the following reaction



Products A and B of above reaction are

(1) $(CH_3)_3C - OH + C_2H_5 - I$

(2) $(CH_3)_3C - I + C_2H_5I$

(3) $(CH_3)_3C - OH + C_2H_5 - OH$

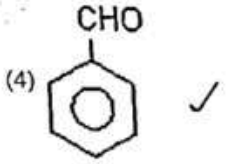
(4) $(CH_3)_3C - I + C_2H_5 - OH$

76. Among the following, which compound will not give Fehling's test?

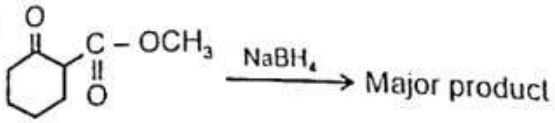
(1) CH_3CHO

(2) HCHO

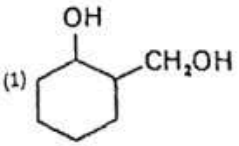
(3) CH_3CH_2CHO

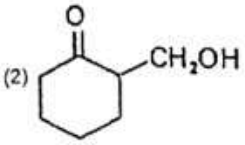
(4) 

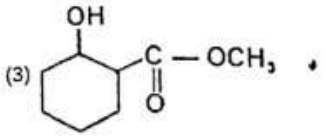
EinsteinX365

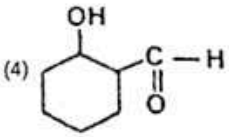
77. 

What is the major product of above reaction?

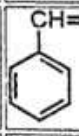
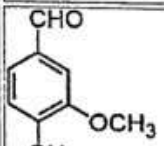
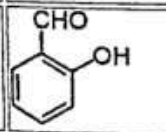
(1) 

(2) 

(3) 

(4) 

78. Match the structures given in **List-I** with their name given in **List-II**.

	List-I		List-II
a.		(i)	Vanillin
b.		(ii)	Acrolein
c.		(iii)	Cinnamaldehyde
d.	$CH_2 = CH - CHO$	(iv)	Salicylaldehyde

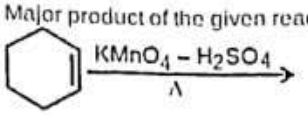
(1) a(iii), b(i), c(ii), d(iv)

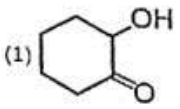
(2) a(iii), b(i), c(iv), d(ii) ✓

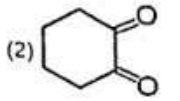
(3) a(i), b(iii), c(iv), d(ii)

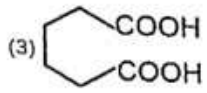
(4) a(ii), b(iv), c(i), d(iii)

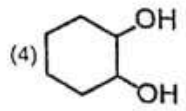
79. Major product of the given reaction is



(1) 

(2) 

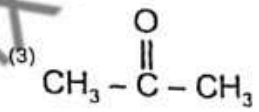
(3)  ✓

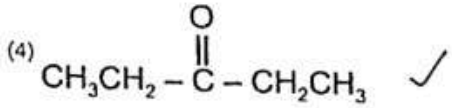
(4) 

80. The compound which will not form iodoform on reaction with $I_2/NaOH$ is

(1) CH_3CHO

(2) CH_3CH_2OH

(3) 

(4)  ✓

81. Which of the following can give carbylamine test?

(1) CH_3CONH_2

(2) $C_2H_5NH_2$ ✓

(3) CH_3NHCH_3

(4) $(CH_3)_3N$

82. Benzene diazonium chloride on treatment with hot water gives

(1) Benzene

(2) Benzyl alcohol

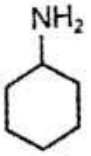
(3) Benzaldehyde

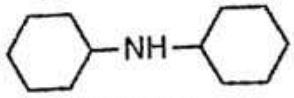
(4) Phenol ✓

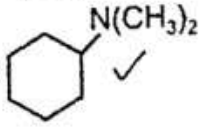
EinsteinX369

83. Which among the following compounds will not react with Hinsberg reagent?

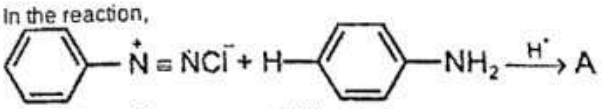
(1) 

(2) 

(3) 

(4) 

84. In the reaction,



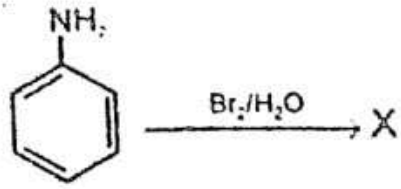
The colour of the compound A is

(1) Orange

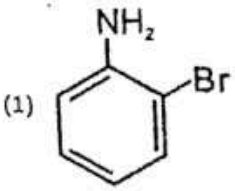
(2) Yellow

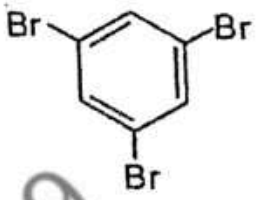
(3) Green

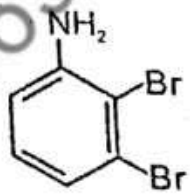
(4) Blue

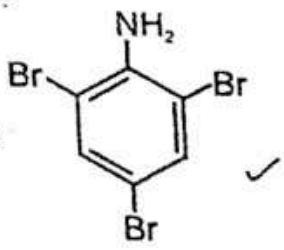
85. 

Major product X is

(1) 

(2) 

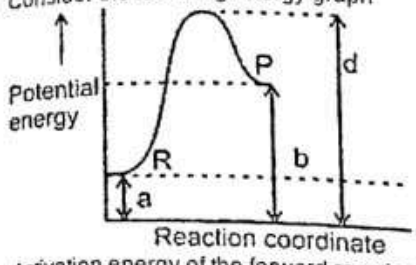
(3) 

(4) 

EinsteinX369

SECTION-B

86. Consider the following energy graph



Activation energy of the forward reaction is

(1) a

(2) a + b

(3) d – a

(4) b – a

87. Incorrect statement about catalyst is

(1) It catalyses forward as well as the back ward reactions to the same extent

(2) It catalyses the spontaneous as well as non-spontaneous reactions

(3) It does not alter Gibbs energy

(4) It does not change the equilibrium constant

88. During the test of halogens, sodium extract is first boiled with nitric acid so as to

(1) Dissolve AgCN

(2) Dissolve Ag_2S

(3) Decompose NaCN and Na_2S

(4) Decompose AgCN and Ag_2S

89. Match List-I with List-II

	List-I		List-II
a.	$Fe_4[Fe(CN)_6]_3.xH_2O$	(i)	Yellow
b.	$(NH_4)_3PO_4.12MoO_3$	(ii)	Prussian Blue
c.	$[Fe(CN)_5NOS]^{4-}$	(iii)	Violet
d.	$[Fe(SCN)]^{2+}$	(iv)	Blood red

Choose the correct answer from the options given below.

(1) a(ii), b(iii), c(iv), d(i)

(2) a(i), b(ii), c(ii), d(iv)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(iii), b(i), c(iv), d(ii)

90. 0.2 g of an organic compound gave 0.188 g of AgBr. The percentage of bromine in the compound is

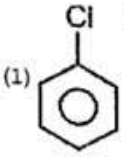
(1) 10%

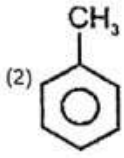
(2) 20%

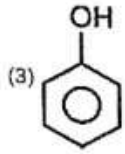
(3) 30%

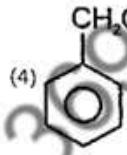
(4) 40%

91. Compound which will give fastest rate of electrophilic aromatic substitution among the following is

(1) 

(2) 

(3) 

(4) 

92. Predict A and B in the following reaction

$$\text{Benzene} \xrightarrow{\Delta} \text{Benzene triozonide} \xrightarrow{B} \text{Glyoxal}$$

(1) A = O_2, B = H_2O

(2) A = O_3, B = Zn/H_2O

(3) A = ZnO, B = O_3/H_2O

(4) A = $KMnO_4/H^+$, B = Zn/H_2O

93. Given below are the two statements.
Statement I : Propan-2-ol is an achiral molecule.
Statement I : Butan-2-ol is a chiral molecule.
In light of the above statements, choose the correct answer

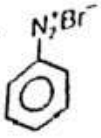
(1) Statement I is correct but statement II is incorrect

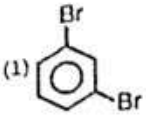
(2) Statement I is incorrect but statement II is correct

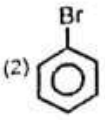
(3) Both statement I and statement II are correct

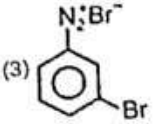
(4) Both statement I and statement II are incorrect

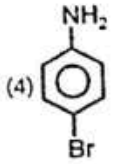
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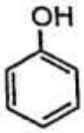
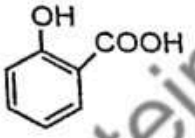
94.  $\xrightarrow{Cu_2Br_2/HBr}$ Product (X). (X) is

(1) 

(2) 

(3) 

(4) 

95.  $\xrightarrow{NaOH}$ (A) $\xrightarrow[(ii)\ H^+]{(i)\ CO_2}$ 

Above reaction is known as

(1) Kolbe's reaction

(2) Reimer-Tiemann reaction

(3) Wurtz reaction

(4) Finkelstein reaction

96. Consider the reaction

$$CH_3 - CN \xrightarrow{H_3O^+} \xrightarrow[\Delta]{NH_3} \xrightarrow{Br_2/KOH} A$$

A is

(1) Ethanamine

(2) Methanamine

(3) Methyl bromide

(4) Ethyl bromide

97. Given below are two statements one is labelled as assertion (A) other is labelled as reason (R)

Assertion (A) : $ClCH_2COOH$ is more acidic than CH_3COOH

Reason (R) : Due to –I effect acidic nature of carboxylic acid decreases.

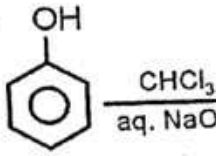
In the light of the above statements choose the correct answer from the options given below

(1) Both A and R are true and R is NOT the correct explanation of A

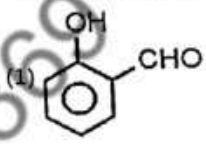
(2) A is true but R is false

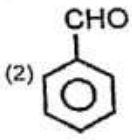
(3) A is false but R is true

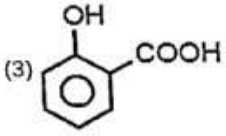
(4) Both A and R are true and R is the correct explanation of A

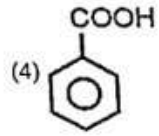
98.  $\xrightarrow[aq.\ NaOH]{CHCl_3}$ [A] $\xrightarrow{NaOH}$ [B] $\xrightarrow{H^+}$ [C]

Product [C] obtained in the above reaction is

(1) 

(2) 

(3) 

(4) 

EinsteinX369

99. Given below are two statements one is labelled as Assertion (A) other is labelled as Reason (R).
Assertion (A) : Aromatic amines are weaker base than ammonia.
Reason (R) : In aromatic amines lone pair of electrons on nitrogen is delocalised in the ring.
In the light of above statements choose the **correct** answer.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

100. The correct order of basicity of given amines in gas phase is

(1) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$

(2) $(C_2H_5)_2NH < (C_2H_5)_3N < C_2H_5NH_2 < NH_3$

(3) $(CH_3)_2NH > (CH_3)_3N > CH_3NH_2 > NH_3$

(4) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2 > NH_3$

EinsteinX369

BOTANY

SECTION-A

101. Select the function of accessory pigments.

(1) Act as primary photosynthetic pigment if chlorophyll a is absent

(2) Trap a wider range of wavelength of light

(3) Protect chlorophyll b only

(4) Absorb only longer wavelength of light spectrum

102. Match the column.

	Column-I (Scientist)		Column-II (Experimental Material)
(a)	T.W Engelmann	(i)	Mint
(b)	Cornelius van Niel	(ii)	*Cladophora*
(c)	Joseph Priestley	(iii)	An aquatic plant
(d)	Jan Ingenhousz	(iv)	Bacteria

(1) a(ii), b(iv), c(i), d(iii)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(i), b(iv), c(iii), d(ii)

(4) a(ii), b(i), c(iv), d(iii)

103. Which of the following is common between cyclic and non-cyclic photophosphorylation?

(1) Synthesis of ATP

(2) Photolysis of water to release O_2

(3) Reduction of $NADP^+$

(4) Accumulation of protons in stroma

104. Which among the given, transfers electrons between PS II and cytochrome b_6f complex?

(1) Plastoquinone

(2) Plastocyanin

(3) Ferredoxin

(4) PS I

105. Glycolysis occurs in the A of the cells. It is defined as the process of B of glucose to form two molecules of pyruvic acid.
Fill the blanks with suitable 'A' and 'B'

	A	B
(1)	Matrix of mitochondria	Complete oxidation
(2)	Cytoplasm	Complete oxidation
(3)	Cytoplasm	Partial oxidation
(4)	Stroma of chloroplast	Partial oxidation

(1) (1)

(2) (2)

(3) (3)

(4) (4)

106. The C_3-plants respond to increased CO_2 concentration and saturation is seen only

(1) At about 360 μlL^{-1}

(2) Beyond 450 μlL^{-1}

(3) Below 360 μlL^{-1}

(4) At about 300 μlL^{-1}

107. Mark the **incorrect** statement(s) w.r.t Z scheme.
(a) Only PS-II is involved
(b) Only PS-I is involved
(c) Splitting of H_2O is associated with PS-I
(d) Chlorophyll *a* in the reaction centre of PS-I has absorption maxima at wavelength 680 nm of the light

(1) Only (a), (b), (c)

(2) Only (a), (b), (d)

(3) All (a), (b), (c) and (d)

(4) Only (d)

108. Respiratory pathway would be better to consider as an amphibolic pathway because

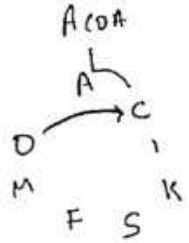
(1) In this pathway, the intermediates do not form any link with breakdown of proteins and fats

(2) Its end products are not any organic molecule

(3) It involves the synthesis of organic products from inorganic substrates

(4) It is involved in both anabolism and catabolism

109. The first step of TCA cycle
a. Starts with condensation of acetyl group with OAA & water
b. Yields a five carbon product
c. Is catalysed by citrate synthase
Choose the **correct** one(s).



(1) a only

(2) a & b only

(3) b & c only

(4) a & c only

110. Total number of ATP molecules produced through oxidative phosphorylation if one molecule of PGAL is completely oxidised during aerobic respiration is

(1) 5

(2) 20

(3) 17

(4) 10

111. Heterophyllous development in buttercup is due to

(1) Variation in temperature

(2) Availability and non-availability of nutrients

(3) Difference in surrounding environment

(4) Presence and absence of light

EinsteinX360

112. Match Column I with II and choose the **correct** option.

	Column I		Column II
a.	Ethylene	(i)	Promotes synthesis of chloroplast
b.	Auxin	(ii)	Promotes bolting
c.	Cytokinin	(iii)	Xylem differentiation
d.	Gibberellin	(iv)	Horizontal growth of seedlings.

(1) a(iv), b(i), c(iii), d(ii)

(2) a(ii), b(i), c(iii), d(iv)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(iv), b(iii), c(i), d(ii)

113. Two leaves (A and B) of a plant have sizes 10 cm^2 and 20 cm^2 respectively. After one day their sizes become 20 cm^2 and 30 cm^2 respectively.
On the basis of given information select the **correct** statement.

(1) Both the leaves show same relative growth rate

(2) Relative growth rate of leaf A is more than that of leaf B

(3) Relative growth rate of leaf B is more than that of leaf A

(4) Absolute growth rate of leaf A is more than that of leaf B

114. α-Thalassemia is

(1) Caused due to abnormality in chromosome 11

(2) Commonly observed in females only

(3) Controlled by two closely linked genes on chromosome 16 of each parent

(4) Controlled by HBA1 gene and HBB gene

115. An exception to Mendelian principle, where both alleles express themselves equally in heterozygous condition is

(1) Incomplete dominance

(2) Codominance

(3) Pleiotropism

(4) Polygenic inheritance

116. In a cross performed by T.H Morgan considering eye colour and wings size in *Drosophila* what is the probability of males having miniature wings in F_2 generation?

(1) 62.8%

(2) 37.2%

(3) 1.3%

(4) 50%

117. In a monohybrid cross, law of dominance is used to explain

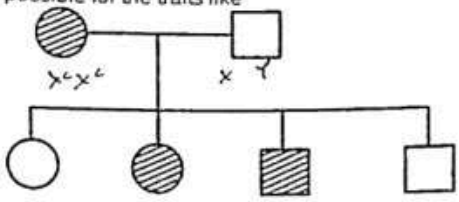
(1) The expression of only one of the parental characters in the F_2 generation

(2) The phenotypic ratio 3:1 obtained in the F_2 generation

(3) The expression of both of the parental characters in the F_1 generation

(4) Purity of gamete.

118. Study the given pedigree chart. This pedigree is **not** possible for the traits like



(1) Haemophilia

(2) Sickle cell anaemia

(3) Cystic fibrosis

(4) Phenylketonuria

119. Haplodiploid method of sex determination is found in

(1) Butterfly

(2) Honey bee

(3) Grasshopper

(4) Fruit fly

120. **Assertion :** In sickle cell anaemia, the shape of RBC changes from biconcave disc to elongated sickle like structure.
Reason : The mutant haemoglobin molecule in sickle cell anaemia undergoes polymerization under low oxygen tension causing the change in shape of RBC.
In the light of the above statements, select the **correct** option.

(1) Both Assertion and Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion and Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

121. How many of the following are dominant traits of pea used by Mendel for his experiments?

Terminal flower, green pod, yellow seed, white flower, constricted pod, green seed

(1) 3

(2) 2

(3) 1

(4) 4

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122. In a double stranded DNA, thymine is 35% of the total nitrogenous bases, then what will be percentage of cytosine?

(1) 35%

(2) 15%

(3) 45%

(4) 20%

123. Two nucleotides are linked through 3' – 5' _____ to form a dinucleotide.

(1) Glycosidic bond

(2) Hydrogen bond

(3) Phosphodiester bond

(4) Peptide bond

124. If a hybrid DNA is allowed to replicate for two generations in N^{14} containing medium then the proportion of light, hybrid and heavy DNA will be respectively

(1) 75%, 25% and 0%

(2) 25%, 25% and 50%

(3) 0%, 50% and 50%

(4) 25%, 0% and 75%

125. Select the **incorrect** statement w.r.t 'bacterial transcription'.

(1) mRNA does not require any processing to become active.

(2) Transcription and translation take place in same compartment as there is no separation of cytosol and nucleus

(3) Translation begins only after the m-RNA is fully transcribed

(4) Transcription and translation can be coupled in bacteria

126. Match the list I with list II w.r.t. *lac* operon and choose the correct option

	List I		List II
a.	*z* gene	(i)	Associated to increase the permeability of the cell to beta galactosides
b.	*y* gene	(ii)	It codes for the enzyme responsible for the hydrolysis of a diasaccharide
c.	*i* gene	(iii)	It encodes for transacetylase
d.	*a* gene	(iv)	Constitutive gene

(1) a(ii), b(i), c(iv), d(iii)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(ii), b(iv), c(iii), d(i)

(4) a(iii), b(ii), c(iv), d(i)

127. In capping, _____ is added to 5' end of

(1) Unusual nucleotide, rRNA

(2) Methyl guanosine triphosphate, hnRNA

(3) Guanosine monophosphate, hnRNA

(4) Unusual nucleotide, tRNA.

128. Three sequential steps of Alfred Hershey and Martha Chase experiments were

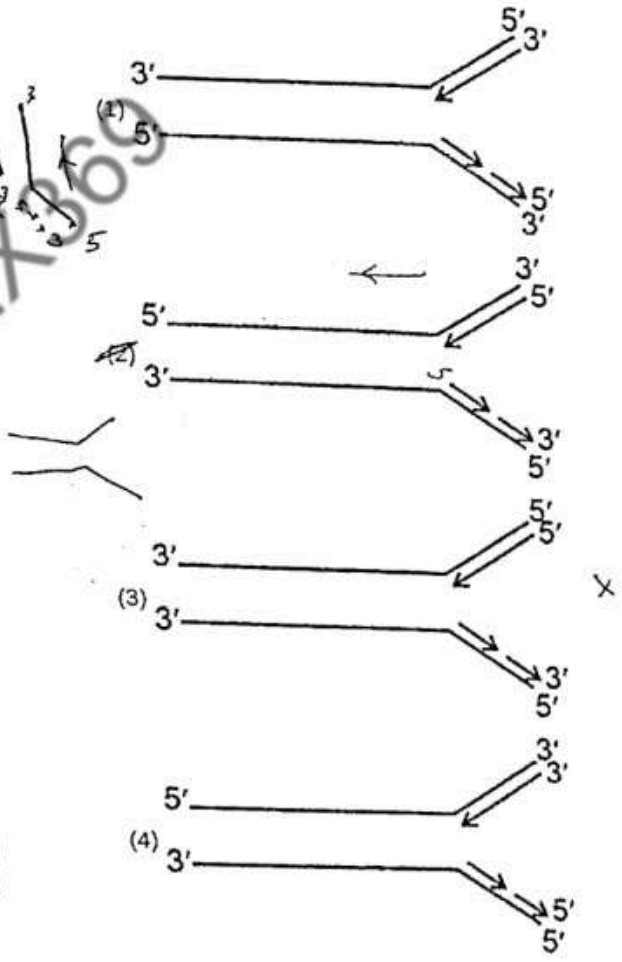
(1) Centrifugation → Blending → Infection

(2) Blending → Centrifugation → Infection

(3) Infection → Blending → Centrifugation

(4) Infection → Centrifugation → Blending

129. Which of the following **correctly** represents the pattern of replication of DNA?



130. Turner's syndrome

(1) Is due to extra copy of chromosome 21

(2) Is due to additional copy of the sex chromosome

(3) Leads to sterile female

(4) Is a Mendelian disorder

EinsteinX369

131. Sporopollenin
a. Helps in fossilization of pollen grains
b. Protects pollen grains from hazardous environment.
c. is present inside the prominent aperture called germ pore.
d. Has no taxonomic significance.
The **correct** one(s) is/are

(1) Only a and b
(2) a, b and d
(3) a and c
(4) Only a

132. The anther wall layer that provides nourishment to developing pollen grains is

(1) Endothecium
(2) Tapetum
(3) Epidermis
(4) Middle layer

133. O. Avery, C. MacLeod and M. McCarty worked to determine the biochemical nature of transforming principle, related to Griffith's experiment. Their work led to conclude that

(1) Transforming substance is RNA
(2) DNA is the hereditary material
(3) Transforming substance is protein
(4) Enzyme proteases and RNases affect transformation

134. Read the following statements and identify them as **true** (T) or **false** (F).
A. Stalk of the ovule is called placenta
B. Ovule is surrounded by integuments
C. Basal part of ovule is the micropyle
D. Ovule is filled with nucellus tissue

	A	B	C	D
(1)	F	T	F	T
(2)	F	T	T	T
(3)	T	F	T	F
(4)	F	F	T	T

(1) (1)
(2) (2)
(3) (3)
(4) (4)

135. Both wind and water pollinated flowers
a. Are not very colourful
b. Do not produce nectar
c. Have feathery stigma
d. Exhibit enormous production of pollen grains
The **correct** ones are

(1) Only a and b
(2) Only c and d
(3) Only a, b and d
(4) All a, b, c and d

SECTION-B

136. Choose the **incorrect** statement w.r.t. the most crucial step of the Calvin cycle.

(1) Is catalysed by the enzyme present in stroma
(2) Occurs even in those plants where fixing of CO_2 into sugar is faciliated by the help of C_4 mechanism
(3) Utilises products of light reaction
(4) Requires CO_2 and 5 C compound as substrate for the most abundant enzyme on earth

137. NADP reductase enzyme is located on the stroma side of thylakoid membrane and accepts

(1) Electrons from PS II and protons from stroma
(2) Electrons from PS II and protons from thylakoid lumen
(3) Electrons from PS I and protons from stroma
(4) Both electrons and protons from stroma

138. In fermentation, reducing agent is

(1) NAD^+
(2) $NADH + H^+$
(3) $NADPH + H^+$
(4) $FADH_2$

139. If one molecule of acetyl CoA undergoes complete oxidation, then how many CO_2 molecules will be released?

(1) One
(2) Two
(3) Three
(4) Four

140. Efficiency index is the measure of

(1) Ability of the plant to produce new plant material
(2) Time of growth
(3) Number of cells that lose capacity to divide
(4) Rate of differentiation of plant cells

Einstein Y365

141. Coconut milk can be used to
(1) Induce root growth and fruit ripening
(2) Promote senescence
(3) Induce formation of male flower
(4) Counteract apical dominance

142. In an angiosperm, total how many haploid nuclei take part in double fertilization?
(1) 3
(2) 5
(3) 4
(4) 2

143. The **correct** sequence of different parts of an ovule from outside to inside is
(1) Nucellus, Integument, Embryo sac
(2) Integument, Nucellus, Embryo sac
(3) Nucellus, Funicle, Chalaza
(4) Integument, Funicle, Nucellus

144. Generative cell of pollen grain
(1) Has large irregularly shaped nucleus
(2) Has dense cytoplasm
(3) Is bigger in size than vegetative cell
(4) Has abundant food reserve

145. In *Antirrhinum*, if a pink flowered plant is self-pollinated then phenotypic and genotypic ratio, will be
(1) 3 : 1
(2) 1 : 2 : 1
(3) 1 : 2 : 1 and 3 : 1, respectively
(4) 3 : 1 and 1 : 2 : 1, respectively

146. Geitonogamy involves
(1) Fertilisation of a flower by the pollen from another flower of the same plant
(2) Fertilization of a flower by the pollen from the same flower
(3) Fertilization of a flower by the pollen from a flower of another plant in the same population
(4) Fertilisation of a flower by the pollen from a flower of another plant belonging to a distant population

147. Human beings have _____ alleles for ABO blood grouping with _____ phenotypes and _____ genotypes
(1) 4, 3, 6
(2) 4, 6, 3
(3) 3, 4, 6
(4) 6, 3, 4

148. Find out the **correctly** matched pair
(1) Translation unit – UTR
(2) Splicing – Represents dominance of RNA world
(3) Monocistronic – Mostly in bacteria
(4) Origin of replication – Several thousands in bacteria

149. **Assertion (A) :** Embryo development precedes endosperm development.
Reason (R) : Early stages of embryogeny are different in both monocots and dicots though seeds are similar.
In the light of the above statements, select the **correct** option
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

150. Gynoecium of *Michelia* is
(1) Monocarpellary syncarpous
(2) Multicarpellary syncarpous
(3) Monocarpellary apocarpous
(4) Multicarpellary apocarpous

ZOOLOGY

SECTION-A

EinsteinX369

151. IUT differs from ZIFT due to

(1) *In vivo* fertilization

(2) *In vitro* fertilization

(3) Transfer of embryo with more than 8 blastomeres into uterus

(4) Transfer of embryo with less than 8 blastomeres into fallopian tube

152. Saheli is the new oral contraceptive with very few side effects. These pills have high contraceptive value and it is a

(1) Non-steroidal pill

(2) Steroidal pill

(3) Once a day pill

(4) Once a month pill

153. Select the set of contraceptive methods which involve a role of hormone.

(1) Lactational amenorrhea, Steroidal pills, Coitus interruptus

(2) Periodic abstinence, Pills, Barrier methods

(3) Implants, Steroidal pills, LNG-20

(4) CuT, Injectables, Emergency contraceptives

154. **Assertion (A) :** After tubectomy, menstrual cycle in a woman stops.
Reason (R) : Tubectomy leads to anovulation.
In the light of above statements, select the **correct** option

(1) Both (A) and (R) are true and (R) is the correct explanation of the (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of the (A)

(3) (A) is true statement but (R) is false

(4) Both (A) and (R) are false statements

155. In lactational amenorrhea, there is absence of ovulation and menstruation during the period of intense lactation following parturition, due to

(1) High levels of estrogen and progesterone

(2) High levels of prolactin, which inhibits the release of gonadotropins

(3) High levels of hCG

(4) High levels of FSH and LH

156. Choose the **incorrect** statement w.r.t. frogs.

(1) Two ureters emerge from the kidneys in males.

(2) In females, the ureter and oviduct open together as the urinogenital duct in the cloaca.

(3) Excretory wastes are carried by blood into the kidney where it is separated and excreted.

(4) Elimination of faecal matter and urine to the exterior occurs through the cloacal aperture.

157. The correct pathway for the passage of sperms in male frogs is

(1) Testes → Vasa efferentia → Kidney → Bidder's canal → Urinogenital duct → Cloaca

(2) Testes → Vasa efferentia → Bidder's canal → Kidney → Cloaca → Ureters

(3) Testes → Bidder's canal → Vasa efferentia → Kidney → Cloaca → Ureters

(4) Testes → Vasa efferentia → Ureters → Kidney → Cloaca → Bidder's canal

158. Male frogs can be differentiated from female frogs by the presence of
a. Sound producing vocal sacs
b. More number of digits in forelimbs
c. Copulatory pads on first digit of forelimbs
d. Webbed digits of the hindlimbs
Select the **correct** option.

(1) a and b

(2) b and c

(3) a and c

(4) a and d

159. Match column I and column II w.r.t. cockroach. Choose the **correct** option.

Column I	Column II
a. Hepatic caecae	(i) 6-8 in number
b. Malpighian tubules	(ii) 100-150 in number
c. Spiracles	(iii) 20 in number
d. Ommatidia in each eye	(iv) 2000 in number

(1) a(ii), b(iii), c(iv), d(i)

(2) a(i), b(ii), c(iii), d(iv)

(3) a(iv), b(ii), c(iii), d(i)

(4) a(iv), b(i), c(ii), d(iii)

160. Male cockroach can be distinguished from the female cockroach by the presence of

(1) Ocelli

(2) Pronotum

(3) Anal style

(4) Anal cerci

161. An adult human male ejaculates 200 - 300 million sperms per ejaculation. For normal fertility, at least what per cent of sperms should have normal shape, size and vigorous motility?

(1) 60

(2) 24

(3) 20

(4) 50

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162. Common site of fertilization and implantation are _____ and _____ respectively.
Choose the option which fill the blanks **correctly.**

(1) Infundibulum, cervix of uterus

(2) Fimbriae, Fundus of uterus

(3) Isthmus, cervix of uterus

(4) Ampulla of fallopian tube, fundus of uterus ✓

163. **Assertion (A):** Inner cell mass carries the potential to differentiate into all the three germ layers.
Reason (R): Inner cell mass contains the stem cells.
In the light of above statements, choose the **correct** answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true and (R) is false

(4) (A) is false and (R) is true

164. Hormones, which stimulate formation of milk and alveoli in mammary glands respectively are

(1) Estrogen, prolactin

(2) Prolactin, progesterone

(3) Progesterone, oxytocin

(4) Oxytocin, Estrogen

165. Difference between oogenesis and spermatogenesis is

(1) Oogenesis begins at puberty and spermatogenesis during embryonic development.

(2) Oogenesis results in formation of four haploid ova from a primary oocyte while during spermatogenesis only one sperm is formed from a primary spermatocyte.

(3) Cytoplasmic division is unequal in oogenesis and equal in spermatogenesis.

(4) Spermatogenesis is not completed until after fertilization occurs, but oogenesis is completed at the time of birth.

166. Estrogens are the dominant ovarian hormones during which phase of menstrual cycle?

(1) Secretory phase

(2) Proliferative phase

(3) Luteal phase

(4) Bleeding phase

167. Spermatogenesis by the seminiferous tubules requires presence of

(1) Oxytocin

(2) Androgens and FSH

(3) Somatostatin

(4) Pitocin

168. Sertoli cells are found in

(1) Primary sex organ of females and secrete estrogen hormone

(2) Primary sex organ of males and secrete androgens

(3) Seminiferous tubules and secrete some factors which help in the process of spermiogenesis

(4) Epididymis and secrete nutritive fluids for sperms

169. In which of the following fishes, air bladder is absent and they have to swim constantly to avoid sinking?

(1) *Exocoetus*

(2) *Labeo*

(3) *Hippocampus*

(4) *Scoliodon*

170. *Aptenodytes* belong to a group of animals having

(1) Mammary glands and 2 pairs of limbs

(2) Epidermal scales with 3-chambered heart

(3) Mixing of oxygenated and deoxygenated blood in ventricle

(4) Oviparity and air sacs to supplement pulmonary respiration

171. Teeth are modified placoid scales and are backwardly directed in

(1) *Trygon*

(2) *Exocoetus*

(3) *Labeo*

(4) *Hippocampus*

172. Choose the **correct** option to complete the given analogy.
Petromyzon :Lamprey : : *Exocoetus* : _____

(1) Hagfish

(2) Saw fish

(3) Flying fish

(4) Sting ray

173. Read the following statements **A** and **B** and choose the **correct** answer.
Statement A : All vertebrates are chordates but all chordates are not vertebrates.
Statement B : All mammals are homeothermous but all homeothermous animals are not mammals.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

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174. How many of the following are the examples of mammals in the box given below?

Testudo, Neophron, Canis, Felis, Pteropus, Catla, Carcharodon, Macropus, Rattus

Choose the **correct** option.

(1) Four
(2) Five
(3) Six
(4) Seven

175. Select the **correct** option that is an ectoparasite on some fishes and their body is devoid of scales and paired fins.

(1) Lamprey
(2) Flying fish
(3) Sea horse
(4) Dog fish

176. Read the following statements A and B and choose the **correct** answer

Statement-A : Members of the phylum Arthropoda are similar to that of Mollusca as both have ventral, solid nerve cord.

Statement-B : Radula is file-like rasping organ for feeding found in molluscs and echinoderms.

(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

177. Which of the following is **correct** set of animals belonging to phylum Mollusca?

(1) *Pila, Loligo, Locusta*
(2) *Loligo, Laccifer, Ophiura*
(3) *Chaetopleura, Aplysia, Sepia*
(4) *Asterias, Octopus, Ophiura*

178. Select the **odd** one w.r.t. sexuality of organism.

(1) Cockroach
(2) Earthworm
(3) Leech
(4) Tapeworm

179. Metamerism is shown by all of the following organisms, **except**

(1) *Ancylostoma*
(2) *Pheretima*
(3) *Laccifer*
(4) *Locusta*

180. The largest and the second largest phylum of animal kingdom respectively are

(1) Chordata and Arthropoda
(2) Porifera and Mollusca
(3) Arthropoda and Mollusca
(4) Annelida and Arthropoda

181. All of the following are features of birds, except

(1) Homeothermous
(2) Four chambered heart
(3) Double circulation
(4) Presence of diaphragm

182. In tunicates, the notochord is present in

(1) Head of adult
(2) Tail of adult
(3) Tail of larva
(4) Head of larva

183. Gills are **not** covered by operculum in

(1) *Exocoetus*
(2) *Scoliodon*
(3) *Labeo*
(4) *Pterophyllum*

184. Find the incorrect option w.r.t class Amphibia

(1) External fertilisation, oviparous
(2) All of them have two pairs of limbs
(3) Skin is moist without scales
(4) Eyes have eyelids

185. Correct location of preen glands in birds is at the base of the

(1) Tail
(2) Beak
(3) Wings
(4) Eyes

SECTION-B

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186. If vas deferens gets blocked, sperms will not reach from

(1) Seminiferous tubules to rete testis

(2) Epididymis to ejaculatory duct

(3) Rete testis to vasa efferentia

(4) Vasa efferentia to epididymis

187. Read the given statements.
Statement I : Restrictions imposed for MTP promote indiscriminate and legalised female foeticide in India.
Statement II : MTP is most widely accepted method of conception in our country.
Select the **correct** option.

(1) Only statement I is true

(2) Only statement II is true

(3) Both statements I and II are true

(4) Both statements I and II are false

188. Formation of second polar body occurs

(1) Before ovulation

(2) After entry of sperm within secondary oocyte

(3) After zygote formation

(4) Before entry of sperm within secondary oocyte

189. **Assertion (A):** The levels of thyroxine, cortisol and progesterone increase during pregnancy.
Reason (R): Increased levels of these hormones support foetal growth, metabolic changes in the mother and maintenance of pregnancy.
In the light of above statements, choose the **correct** answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

190. The contraceptive device inserted by the doctors or expert nurses in the uterus through vagina is

(1) Implant

(2) IUD

(3) Cervical cap

(4) Diaphragm

191. Given below is a list of animals.
(a) *Pila*
(b) *Ascaris*
(c) *Ascidia*
(d) *Octopus*
(e) *Nereis*
Identify those animals in which adult stage are bilaterally symmetrical, triploblastic and have true coelom.
Choose the **correct** option.

(1) (a) & (b)

(2) (c), (d) & (e)

(3) (a), (d) & (e)

(4) (a), (b), (d) & (e)

192. The larva of which animal undergoes metamorphosis in fresh water and then returns to the ocean?

(1) *Petromyzon*

(2) *Branchiostoma*

(3) *Ascidia*

(4) *Salpa*

193. Closed type of circulation and 6-15 pairs of gill slits for respiration are present in the members of

(1) Cyclostomata

(2) Urochordata

(3) Cephalochordata

(4) Hemichordata

194. Select the soft bodied organism with distinct mantle among the following.

(1) Sea lily

(2) Sea fan

(3) Sea urchin

(4) Sea hare

195. Which of the following wings of cockroach contribute in flying?

(1) Forewings

(2) Tegmina

(3) Metathoracic wings

(4) Mesothoracic wings

196. If in a female, menstrual cycle is of 45 days then the fertile period in which couple should avoid coitus to prevent pregnancy is from

(1) 10th - 17th day

(2) 30th - 37th day

(3) 27th - 34th day

(4) 40th - 47th day

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197. In a human female, how many primary oocytes are required for the formation of 400 ova?

(1) 200

(2) 400

(3) 100

(4) 50

198. The step at which first meiotic division takes place during spermatogenesis is

(1) Spermatogonia → Primary spermatocytes

(2) Primary spermatocytes → Secondary spermatocytes

(3) Secondary spermatocytes → Spermatids

(4) Spermatids → Spermatogonia

199. Among the following given options, which one is not a function of water vascular system in echinoderms?

(1) Locomotion

(2) Respiration

(3) Food capturing

(4) Regeneration

200. Adult "X" are radially symmetrical whereas their larvae are bilaterally symmetrical. Identify "X" from the options given below.

(1) Echinoderms

(2) Poriferans

(3) Coelenterates

(4) Hemichordates

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